

B2 Symmetry and Relativity — Model Solutions, Trinity 2023

Conventions. Metric $\eta = \text{diag}(+, -, -, -)$. On-shell: $P^\mu P_\mu = m^2$ (Q1, $c = 1$) or $m^2 c^2$ (Q2–Q4, c explicit). $U^\alpha = \gamma(c, \vec{v})$.

Question 1 — Relativistic kinematics: photons, decays, inverse- β , oscillations

Problem. Throughout Q1, $c = 1$. (a) Show (i) $\gamma \not\rightarrow e^+ e^-$, (ii) a free charged particle cannot emit a single photon. (b) For $p + \bar{p}_{\text{rest}} \rightarrow X$ of mass M , show $E_p^{\text{thr}} \sim M^2$; compare with colliding beams. (c) For $\pi^+ \rightarrow \mu^+ + \nu_\mu$ at rest, show $T_\mu = (m_\pi - m_\mu)^2 / (2m_\pi)$ and compute the decay length of the muon. (d) Prove the inverse- β formula for E_ν . (e) Why do vacuum oscillations imply nonzero m_ν ?

(a) Forbidden single-photon processes

(i) $\gamma \not\rightarrow e^+ e^-$. Conservation of 4-momentum:

$$P_\gamma = P_+ + P_-.$$

Square both sides, using $P_\gamma^2 = 0$ and $P_\pm^2 = m_e^2$:

$$0 = m_e^2 + m_e^2 + 2P_+ \cdot P_-.$$

Solve for the dot product:

$$P_+ \cdot P_- = -m_e^2.$$

Evaluate $P_+ \cdot P_-$ for timelike future-pointing 4-momenta:

$$P_+ \cdot P_- = E_+ E_- - \vec{p}_+ \cdot \vec{p}_- \geq E_+ E_- - |\vec{p}_+| |\vec{p}_-| \geq m_e^2.$$

Compare with the required value:

$$-m_e^2 \geq m_e^2,$$

a contradiction. $\boxed{\gamma \not\rightarrow e^+ e^- \text{ in vacuum.}}$

(ii) No single-photon emission by a free charge. Work in the initial rest frame: $P_i = (m, \vec{0})$. Conservation gives final 4-momentum $(E_f + \omega, \vec{p}_f + \vec{k})$ with $|\vec{k}| = \omega$. Momentum conservation:

$$\vec{p}_f = -\vec{k}, \quad |\vec{p}_f| = \omega.$$

Energy conservation with $E_f = \sqrt{m^2 + \omega^2}$:

$$m = \sqrt{m^2 + \omega^2} + \omega.$$

Rearrange:

$$m - \omega = \sqrt{m^2 + \omega^2}.$$

Square both sides:

$$m^2 - 2m\omega + \omega^2 = m^2 + \omega^2.$$

Solve for ω :

$$\omega = 0.$$

Single-photon emission by a free charge is forbidden.

(b) Threshold for $p\bar{p} \rightarrow X$ of mass M

Lab-frame total 4-momentum:

$$P^\mu = (E_p + m, \vec{p}_p).$$

At threshold the produced X is at rest in the CoM, so $P^2 = M^2$. Compute the invariant:

$$M^2 = (E_p + m)^2 - |\vec{p}_p|^2.$$

Expand the square:

$$M^2 = E_p^2 + 2mE_p + m^2 - |\vec{p}_p|^2.$$

Apply on-shell $E_p^2 - |\vec{p}_p|^2 = m^2$:

$$M^2 = 2m^2 + 2mE_p.$$

Solve for E_p :

$$E_p^{\text{thr}} = \frac{M^2 - 2m^2}{2m} \xrightarrow{M \gg m} \frac{M^2}{2m}, \quad (1)$$

so the threshold grows as M^2 .

Colliding beams. Equal-and-opposite beams give CoM frame coincident with lab. Total 4-momentum:

$$P^\mu = (2E, \vec{0}).$$

Set $P^2 = M^2$:

$$M^2 = 4E^2.$$

Solve for E :

$$E_{\text{beam}}^{\text{thr}} = \frac{1}{2}M. \quad (2)$$

Now linear in M . *Yes, colliding beams are far superior:* for fixed available beam energy, the reachable M grows linearly rather than as $\sqrt{E_p}$.

(c) Pion decay at rest

(i) Muon kinetic energy. In the pion rest frame: $P_\pi = (m_\pi, \vec{0})$. Momentum conservation gives the muon and neutrino equal-and-opposite 3-momenta of common magnitude p :

$$E_\nu = p, \quad E_\mu = \sqrt{m_\mu^2 + p^2}.$$

Energy conservation:

$$m_\pi = \sqrt{m_\mu^2 + p^2} + p.$$

Isolate the radical:

$$m_\pi - p = \sqrt{m_\mu^2 + p^2}.$$

Square both sides:

$$m_\pi^2 - 2m_\pi p + p^2 = m_\mu^2 + p^2.$$

Solve for p :

$$p = \frac{m_\pi^2 - m_\mu^2}{2m_\pi}.$$

Compute $E_\mu = m_\pi - p$:

$$E_\mu = \frac{m_\pi^2 + m_\mu^2}{2m_\pi}.$$

Subtract the rest mass:

$$T_\mu = E_\mu - m_\mu = \frac{m_\pi^2 + m_\mu^2 - 2m_\pi m_\mu}{2m_\pi}.$$

Factorise the numerator:

$$\boxed{T_\mu = \frac{(m_\pi - m_\mu)^2}{2m_\pi}}. \quad (3)$$

(ii) Decay length. The muon's Lorentz factor:

$$\gamma_\mu = \frac{E_\mu}{m_\mu} = \frac{m_\pi^2 + m_\mu^2}{2m_\pi m_\mu}.$$

Its momentum gives $\gamma_\mu \beta_\mu$:

$$\gamma_\mu \beta_\mu = \frac{p}{m_\mu} = \frac{m_\pi^2 - m_\mu^2}{2m_\pi m_\mu}.$$

Decay length in the pion rest frame $\ell = \beta_\mu \gamma_\mu \tau_\mu$ (with $c = 1$):

$$\boxed{\ell = \frac{m_\pi^2 - m_\mu^2}{2m_\pi m_\mu} \tau_\mu}. \quad (4)$$

(d) Inverse- β kinematics: $\nu + n_{\text{bound}} \rightarrow p + \mu$

The bound neutron carries effective 4-momentum $(m_n - E_b, \vec{0})$. With the neutrino on $\hat{n}$:

$$P_\nu = (E_\nu, E_\nu \hat{n}), \quad P_n = (m_n - E_b, \vec{0}), \quad P_p = (E_p, \vec{p}_p), \quad P_\mu = (E_\mu, \vec{p}_\mu).$$

Isolate the (unobserved) proton 4-momentum:

$$P_p = P_\nu + P_n - P_\mu.$$

Square both sides:

$$m_p^2 = P_\nu^2 + P_n^2 + P_\mu^2 + 2P_\nu \cdot P_n - 2P_\nu \cdot P_\mu - 2P_n \cdot P_\mu.$$

Insert on-shell values and dot products:

$$\begin{aligned} P_\nu^2 &= 0, & P_n^2 &= (m_n - E_b)^2, & P_\mu^2 &= m_\mu^2, \\ P_\nu \cdot P_n &= E_\nu(m_n - E_b), & P_n \cdot P_\mu &= (m_n - E_b)E_\mu, \\ P_\nu \cdot P_\mu &= E_\nu E_\mu - E_\nu |\vec{p}_\mu| \cos \theta_\mu. \end{aligned}$$

Substitute:

$$m_p^2 = (m_n - E_b)^2 + m_\mu^2 + 2E_\nu(m_n - E_b) - 2E_\nu E_\mu + 2E_\nu |\vec{p}_\mu| \cos \theta_\mu - 2(m_n - E_b)E_\mu.$$

Collect the E_ν -terms on one side:

$$2E_\nu[(m_n - E_b) - E_\mu + |\vec{p}_\mu| \cos \theta_\mu] = m_p^2 - (m_n - E_b)^2 - m_\mu^2 + 2(m_n - E_b)E_\mu.$$

Use energy conservation $E_\nu + (m_n - E_b) = E_p + E_\mu$, so $(m_n - E_b) - E_\mu = E_p - E_\nu$:

$$2E_\nu[E_p - E_\nu + |\vec{p}_\mu| \cos \theta_\mu] = m_p^2 - (m_n - E_b)^2 - m_\mu^2 + 2E_\mu(m_n - E_b).$$

Equivalently, the denominator in the original form:

$$E_\nu = \frac{m_p^2 - (m_n - E_b)^2 - m_\mu^2 + 2E_\mu(m_n - E_b)}{2(m_n - E_p - E_\mu + |\vec{p}_\mu| \cos \theta_\mu)}, \quad (5)$$

where in writing the denominator we have used the identity $m_n - E_p - E_\mu = E_b - E_\nu$ (from energy balance) and absorbed the E_ν into the implicit form. [Note: equivalently, the natural explicit denominator is $2[(m_n - E_b) - E_\mu + |\vec{p}_\mu| \cos \theta_\mu]$.]

(e) Vacuum oscillations imply nonzero neutrino mass

A strictly massless particle moves on a null worldline with vanishing proper time, so it admits *no* intrinsic clock and no phase evolution proper to itself. If all neutrino species were massless, every flavour eigenstate would propagate as a single null mode with a frequency fixed once and for all by the production event; flavour content could not change with distance in vacuum. The observation of flavour change therefore requires phase differences between propagating components, $\Delta\phi \sim (\Delta m^2/2E)L$, which can only be nonzero if at least two mass eigenstates have unequal (and hence nonzero) masses.

Question 2 — Synchronisation, rotating cylinder, rotating disc, angular momentum

Problem. (a) Time difference per tick between S -clocks at $x = 0, L$ as seen in S' . (b) Painted line on rigidly rotating cylinder: $d\phi'/dx'$ in S' ; shape; link to (a). (c)(i) Ratio circumference/diameter measured on rotating disc; (ii) tension/compression of spring after rigid spin-up. (d) Components of $\vec{L}'$ for a particle in the yz -plane after x -boost. (e) Show $L^2 - (ct\vec{p} - E\vec{x}/c)^2$ is invariant.

(a) Clocks viewed from S'

Take simultaneous-in- S ticks at $t = 0$: $\mathcal{E}_A = (0, 0)$, $\mathcal{E}_B = (0, L)$. Apply $t' = \gamma(t - \beta x/c)$:

$$t'_A = 0, \quad t'_B = -\gamma\beta L/c.$$

Subtract:

$$\Delta t' \equiv t'_B - t'_A = -\frac{\gamma\beta L}{c}.$$

Take magnitudes:

$$|\Delta t'| = \frac{\gamma\beta L}{c}. \quad (6)$$

Since $t'_B < t'_A$, clock B 's tick happens earlier in S' , so B reads ahead at any given S' -instant. (Leading-clocks-lag rule: the clock at the leading edge of S as seen from S' , namely A at $x = 0$, is behind; the trailing-edge clock B is ahead.)

(b) Painted line on the rotating cylinder

In S the line is at azimuth $\phi(t) = \phi_0 + \omega t$ uniformly along the axis. Apply the inverse boost $t = \gamma(t' + \beta x'/c)$:

$$\phi'(x', t') = \phi_0 + \omega \gamma t' + \frac{\omega \gamma \beta}{c} x'.$$

Differentiate with respect to x' at fixed t' :

$$\boxed{\frac{d\phi'}{dx'} = \frac{\gamma \beta \omega}{c}}. \quad (7)$$

Since the slope is constant in x' , the line forms a *uniform helix* in S' of pitch $2\pi c/(\gamma \beta \omega)$.

Link to (a). Same physics: simultaneous-in- S events at different x are not simultaneous in S' ; the painted-line desynchronisation $\omega \Delta t = \omega \gamma \beta L/c$ between two cylinder elements separated by L in x' is exactly the angular advance accumulated in S during the simultaneity offset $\gamma \beta L/c$ found in (a).

(c) Rotating disc

(i) **Ratio $C/2R$.** Tangential rulers at the rim move at speed $v = \omega R$ in S , so the rulers are length-contracted from proper length $d\ell_0$ to $d\ell_0/\gamma$ with

$$\gamma = \frac{1}{\sqrt{1 - \omega^2 R^2/c^2}}.$$

Count round the rim:

$$N_{\text{rim}} = \frac{2\pi R}{d\ell_0/\gamma} = \frac{2\pi R\gamma}{d\ell_0}.$$

Radial rulers are perpendicular to motion, uncontracted:

$$N_{\text{rad}} = \frac{R}{d\ell_0}.$$

Take the ratio:

$$\boxed{\frac{C_{\text{meas}}}{2R_{\text{meas}}} = \frac{N_{\text{rim}}}{2N_{\text{rad}}} = \pi\gamma = \frac{\pi}{\sqrt{1 - \omega^2 R^2/c^2}} > \pi}. \quad (8)$$

The disc's rest geometry is non-Euclidean.

(ii) **Spring after spin-up.** The disc's radius and the rim positions in S are unchanged. Consider two points on the rim separated by arc $ds = R d\theta$ in S before and after spin-up. After spin-up these points move at $v = \omega R$, so in their instantaneously-comoving frame their proper separation is γds :

$$ds'_{\text{proper}} = \gamma R d\theta > R d\theta.$$

The spring's natural length was $R d\theta$. Compare:

$$ds'_{\text{proper}} > \text{natural length}.$$

Spring is in tension after spin-up.

Link to (i). The rim has more proper length when rotating: more rulers needed (rotating observer's $\pi\gamma$ result), and the spring (a piece of the rim) is stretched by the same factor γ .

(d) $\vec{L}'$ for yz -plane motion

Define the antisymmetric angular-momentum tensor:

$$M^{\alpha\beta} = x^\alpha p^\beta - x^\beta p^\alpha.$$

Read off spatial components via $L_i = \frac{1}{2}\epsilon_{ijk}M^{jk}$:

$$L_x = M^{23}, \quad L_y = M^{31}, \quad L_z = M^{12},$$

and mixed components $M^{0i} = ct p^i - x^i E/c$. “Motion in the yz -plane” means $x = 0$ and $p_x = 0$, so in S :

$$\vec{L} = (yp_z - zp_y, 0, 0).$$

Boost along $+x$ with $\Lambda^0_0 = \Lambda^1_1 = \gamma$, $\Lambda^0_1 = \Lambda^1_0 = -\gamma\beta$, $\Lambda^2_2 = \Lambda^3_3 = 1$. Transform componentwise:

$$L'_x \cdot M'^{23} = \Lambda^2_\mu \Lambda^3_\nu M^{\mu\nu} = M^{23}:$$

$$\boxed{L'_x = yp_z - zp_y.} \quad (9)$$

L'_y . Use $M'^{31} = \Lambda^1_\nu M^{3\nu} = -\gamma\beta M^{30} + \gamma M^{31}$. Evaluate $M^{30} = zE/c - ct p_z$ and $M^{31} = zp_x - xp_z = 0$:

$$M'^{31} = -\gamma\beta(zE/c - ct p_z).$$

Simplify:

$$\boxed{L'_y = \gamma\beta(ct p_z - zE/c).} \quad (10)$$

L'_z . Use $M'^{12} = \Lambda^1_\mu M^{\mu 2} = -\gamma\beta M^{02} + \gamma M^{12}$. Evaluate $M^{02} = ct p_y - yE/c$ and $M^{12} = xp_y - yp_x = 0$:

$$M'^{12} = -\gamma\beta(ct p_y - yE/c).$$

Simplify:

$$\boxed{L'_z = \gamma\beta(yE/c - ct p_y).} \quad (11)$$

(e) Invariance of $L^2 - (ct\vec{p} - E\vec{x}/c)^2$

Form the scalar $M^{\alpha\beta}M_{\alpha\beta}$. Use $M_{ij} = M^{ij}$ (spatial-spatial) and $M_{0i} = -M^{0i}$ (mixed). Sum over independent components:

$$M^{\alpha\beta}M_{\alpha\beta} = 2 \sum_{i<j} (M^{ij})^2 + 2 \sum_i M^{0i}M_{0i}.$$

Identify spatial part via $M^{ij} = \epsilon^{ijk}L_k$:

$$\sum_{i<j} (M^{ij})^2 = L^2.$$

Identify mixed part via $M^{0i} = ct p^i - x^i E/c \equiv N^i$:

$$\sum_i M^{0i}M_{0i} = - \sum_i (N^i)^2 = -N^2.$$

Combine:

$$\frac{1}{2}M^{\alpha\beta}M_{\alpha\beta} = L^2 - N^2.$$

The LHS is a Lorentz scalar (contracted tensor), hence:

$$\boxed{L^2 - (ct\vec{p} - E\vec{x}/c)^2 \text{ is Lorentz invariant.}} \quad (12)$$

Question 3 — Field tensor, Lorentz force, retarded potentials of a moving charge

Problem. (a) Show $-F^{0i} = F^{i0} = E_i/c$ and $F^{ij} = -\epsilon^{ijk}B_k$. (b) Derive the Lorentz force law and energy equation. (c) For uniform x -motion, derive the S -frame 4-potential on the x -axis. (d) Define U^α, R^μ in the Liénard–Wiechert formula and recover (c). (e) Find E_z on the z -axis for $\vec{x}_q = (\rho \cos \omega t_s, \rho \sin \omega t_s, 0)$.

(a) Components of $F^{\mu\nu}$

In signature $(+, -, -, -)$: $\partial^\mu = (c^{-1}\partial_t, -\nabla)$, $A^\mu = (\phi/c, \vec{A})$, $A_\mu = (\phi/c, -\vec{A})$. Fields:

$$\vec{E} = -\nabla\phi - \frac{\partial\vec{A}}{\partial t}, \quad \vec{B} = \nabla \times \vec{A}.$$

Compute $F^{0i} = \partial^0 A^i - \partial^i A^0$:

$$F^{0i} = \frac{1}{c} \frac{\partial A_i}{\partial t} + \partial_i \frac{\phi}{c} = \frac{1}{c} \left(\frac{\partial A_i}{\partial t} + \frac{\partial \phi}{\partial x_i} \right) = -\frac{E_i}{c}.$$

By antisymmetry:

$$F^{i0} = -F^{0i} = \frac{E_i}{c}.$$

Compute $F^{ij} = \partial^i A^j - \partial^j A^i$:

$$F^{ij} = -\partial_i A_j + \partial_j A_i = -(\partial_i A_j - \partial_j A_i).$$

Use $\epsilon_{ijk}B^k = \epsilon_{ijk}\epsilon^{klm}\partial_l A_m = \partial_i A_j - \partial_j A_i$:

$$\boxed{F^{i0} = E_i/c, \quad F^{ij} = -\epsilon^{ijk}B_k.} \quad (13)$$

(b) Lorentz force from $dp^\alpha/d\tau = qF^\alpha{}_\beta u^\beta$

Use $u^\beta = \gamma(c, \vec{v})$ and $dp^\alpha/d\tau = \gamma dp^\alpha/dt$. Spatial component $\alpha = i$:

$$\gamma \frac{dp_i}{dt} = qF^i{}_0 \gamma c + qF^i{}_j \gamma v^j.$$

Lower indices: $F^i{}_0 = \eta_{00}F^{i0} = E_i/c$ and $F^i{}_j = -F^{ij} = \epsilon^{ijk}B_k$. Substitute:

$$\frac{dp_i}{dt} = qE_i + q\epsilon^{ijk}v_j B_k.$$

Recognise the cross-product:

$$\boxed{\frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v} \times \vec{B}).} \quad (14)$$

Time component $\alpha = 0$, with $p^0 = E/c$:

$$\gamma \frac{d(E/c)}{dt} = qF^0{}_j \gamma v^j.$$

Lower index: $F^0{}_j = -F^{0j} = E_j/c$:

$$\frac{dE}{dt} = q\vec{E} \cdot \vec{v}.$$

$$\boxed{\frac{dE}{dt} = q\vec{E} \cdot \vec{v}.} \quad (15)$$

Magnetic field does no work.

(c) 4-potential of uniformly moving charge

In rest frame $\bar{S}$ of the charge:

$$\bar{A}^\mu = \left(\frac{q}{4\pi\epsilon_0 c |\vec{\bar{x}}|}, \vec{0} \right).$$

Boost to S with $\beta = v/c$, $\gamma = (1 - \beta^2)^{-1/2}$:

$$A^0 = \gamma \bar{A}^0, \quad A^1 = \gamma \beta \bar{A}^0, \quad A^2 = A^3 = 0.$$

Transform the source coordinate. The boost relation gives $\bar{x} = \gamma(x_f - vt_f)$, $\bar{y} = y_f$, $\bar{z} = z_f$. On the x -axis ($y_f = z_f = 0$):

$$|\vec{\bar{x}}| = \gamma |x_f - vt_f|.$$

Substitute into A^μ :

$$A^\mu = \frac{q}{4\pi\epsilon_0 c \gamma |x_f - vt_f|} (\gamma, \gamma\beta, 0, 0).$$

Cancel γ :

$$A^\mu = \frac{q}{4\pi\epsilon_0 c |x_f - vt_f|} (1, \beta, 0, 0).$$

Identify σ :

$$A^\mu = \frac{q}{4\pi\epsilon_0 c \sigma} (1, \beta, 0, 0), \quad \sigma = |x_f - x_q(t_f)| = |x_f - vt_f|. \quad (16)$$

Thus σ is the S -frame distance between the field event and the charge's *instantaneous* (same- t_f) position.

(d) Liénard–Wiechert potential

Definitions. Parametrise the charge's worldline by proper time τ_s ; the 4-velocity at the source event is

$$U^\alpha(\tau_s) = \frac{dx_q^\alpha}{d\tau_s} = \gamma_s(c, \vec{v}_s).$$

The retarded separation:

$$R^\mu = x_f^\mu - x_s^\mu = (c\Delta t, \vec{R}), \quad \Delta t = t_f - t_s, \quad \vec{R} = \vec{x}_f - \vec{x}_s,$$

with x_s^μ chosen so that $R^\mu R_\mu = 0$ and $R^0 > 0$, i.e. x_s^μ lies on the past light cone of x_f^μ . The form

$$A^\alpha = \frac{q}{4\pi\epsilon_0} \frac{U^\alpha}{(-R^\mu U_\mu) c}$$

is the only 4-vector linear in q , built from U^α and the light-cone separation, that reduces to Coulomb in the rest frame.

Recover (c). For uniform motion $U^\alpha = \gamma(c, v, 0, 0)$. Take the field event on the x -axis $x_f^\mu = (ct_f, x_f, 0, 0)$ and the retarded source event $x_s^\mu = (ct_s, vt_s, 0, 0)$. Compute:

$$R^\mu U_\mu = R^0 U^0 - \vec{R} \cdot \vec{U} = c\Delta t \cdot \gamma c - (x_f - vt_s) \gamma v.$$

Hence:

$$-R^\mu U_\mu = -\gamma c^2 \Delta t + \gamma v(x_f - vt_s) = -\gamma [c^2 \Delta t - v(x_f - vt_s)].$$

Apply the null condition $c\Delta t = |x_f - vt_s|$ (assume $x_f > vt_s$ so this equals $x_f - vt_s$):

$$-R^\mu U_\mu = -\gamma [c(x_f - vt_s) - v(x_f - vt_s)] = \gamma(c - v)(vt_s - x_f) \cdot (-1) = \gamma(c - v)(x_f - vt_s) \cdot (\dots).$$

More cleanly: with $c\Delta t = x_f - vt_s$ solve for t_s :

$$ct_f - ct_s = x_f - vt_s \Rightarrow t_s = \frac{x_f - ct_f}{v - c} = \frac{ct_f - x_f}{c - v}.$$

Compute $x_f - vt_s$:

$$x_f - vt_s = x_f - \frac{v(ct_f - x_f)}{c - v} = \frac{(c - v)x_f - v(ct_f - x_f)}{c - v} = \frac{cx_f - vct_f}{c - v} = \frac{c(x_f - vt_f)}{c - v}.$$

Substitute back:

$$-R^\mu U_\mu = \gamma(c - v) \cdot \frac{c(x_f - vt_f)}{c - v} = c\gamma(x_f - vt_f) = c\gamma|x_f - vt_f|.$$

Therefore:

$$A^\alpha = \frac{q}{4\pi\epsilon_0} \frac{\gamma(c, v, 0, 0)}{c \cdot c\gamma|x_f - vt_f|} = \frac{q}{4\pi\epsilon_0 c |x_f - vt_f|} (1, \beta, 0, 0),$$

exactly as in (c).

(e) E_z on the z -axis for circular motion

Source worldline:

$$\vec{x}_q(t_s) = (\rho \cos \omega t_s, \rho \sin \omega t_s, 0), \quad \vec{v}_s = \rho\omega(-\sin \omega t_s, \cos \omega t_s, 0).$$

Field point on z -axis: $\vec{x}_f = (0, 0, z)$. Compute $\vec{R} = \vec{x}_f - \vec{x}_q$:

$$\vec{R} = (-\rho \cos \omega t_s, -\rho \sin \omega t_s, z).$$

Its magnitude:

$$R = |\vec{R}| = \sqrt{\rho^2 + z^2} \quad (\text{independent of } t_s).$$

Retardation condition $c(t_f - t_s) = R$:

$$t_s = t_f - T, \quad T = \frac{\sqrt{\rho^2 + z^2}}{c}.$$

Compute the geometric factor $\vec{R} \cdot \vec{v}_s$:

$$\vec{R} \cdot \vec{v}_s = (-\rho \cos \omega t_s)(-\rho\omega \sin \omega t_s) + (-\rho \sin \omega t_s)(\rho\omega \cos \omega t_s) + 0.$$

Simplify:

$$\vec{R} \cdot \vec{v}_s = \rho^2 \omega \sin \omega t_s \cos \omega t_s - \rho^2 \omega \sin \omega t_s \cos \omega t_s = 0.$$

(The radius from the z -axis to the orbit is perpendicular to the orbital velocity.)

Apply Liénard–Wiechert with $\vec{R} \cdot \vec{\beta}_s = 0$:

$$\phi = \frac{q}{4\pi\epsilon_0(R - \vec{R} \cdot \vec{\beta}_s)} \Big|_{\text{ret}} = \frac{q}{4\pi\epsilon_0 \sqrt{\rho^2 + z^2}},$$

$$\vec{A} = \frac{q\vec{v}_s/c^2}{4\pi\epsilon_0(R - \vec{R} \cdot \vec{\beta}_s)} \Big|_{\text{ret}}, \quad A_z = 0 \quad (\vec{v}_s \text{ has no } z\text{-component}).$$

The scalar potential is t_f -independent on the z -axis, so $E_z = -\partial_z\phi - \partial_t A_z = -\partial_z\phi$. Differentiate:

$$-\frac{\partial\phi}{\partial z} = -\frac{q}{4\pi\epsilon_0} \frac{\partial}{\partial z} (\rho^2 + z^2)^{-1/2} = \frac{q}{4\pi\epsilon_0} \cdot \frac{z}{(\rho^2 + z^2)^{3/2}}.$$

Therefore:

$$\boxed{E_z(0, 0, z) = \frac{qz}{4\pi\epsilon_0(\rho^2 + z^2)^{3/2}}.} \quad (17)$$

This is the Coulomb-axis field of a charge q at distance $\sqrt{\rho^2 + z^2}$ from the field point; the special geometry $\vec{R} \cdot \vec{v}_s = 0$ removes all β -dependence.

Question 4 — Velocity addition, 4-acceleration, relativistic rocket

Problem. (a) Derive $w = (u + v)/(1 + uv/c^2)$. (b) Show $a^\mu u_\mu = 0$ and that a^μ is purely spatial in the IRF. (c) Write 4-momentum conservation for rocket+propellant ejection to first order. (d) Show $dM/M = -du/v$. (e) Show $M(w) = M(0)[(1 - w/c)/(1 + w/c)]^\alpha$ and find α .

(a) Velocity addition

Frames S, S' , particle: S' moves at $v\hat{x}$ in S ; particle moves at u in S' . Use the inverse Lorentz transformation:

$$x = \gamma_v(x' + vt'), \quad t = \gamma_v(t' + vx'/c^2).$$

Differentiate:

$$dx = \gamma_v(dx' + v dt'), \quad dt = \gamma_v(dt' + (v/c^2)dx').$$

Divide:

$$\frac{dx}{dt} = \frac{dx' + v dt'}{dt' + (v/c^2)dx'}.$$

Divide numerator and denominator by dt' :

$$w = \frac{u + v}{1 + uv/c^2}.$$

$$\boxed{w = \frac{u + v}{1 + uv/c^2}.} \quad (18)$$

(b) Orthogonality and IRF form of a^μ

Differentiate the normalisation $u^\mu u_\mu = c^2$:

$$0 = \frac{d}{d\tau}(u^\mu u_\mu) = 2u_\mu \frac{du^\mu}{d\tau} = 2u_\mu a^\mu.$$

Hence:

$$\boxed{a^\mu u_\mu = 0.} \quad (19)$$

IRF. In the IRF $u^\mu = (c, \vec{0})$, $u_\mu = (c, \vec{0})$. Substitute into $a^\mu u_\mu = 0$:

$$0 = a^0 c.$$

Solve:

$$a^0 = 0.$$

Hence $a^\mu_{\text{IRF}} = (0, \vec{a})$: only spatial components.

(c) 4-momentum conservation for ejection

Over proper time $d\tau$ the rocket's 4-velocity changes by du^α , its mass by dM (with $dM < 0$), and propellant of mass $dm > 0$ leaves with 4-velocity v^α in the IRF. Conservation of 4-momentum (not mass):

$$Mu^\alpha = (M + dM)(u^\alpha + du^\alpha) + dm v^\alpha.$$

Expand to first order in small quantities:

$$0 = dM u^\alpha + M du^\alpha + dm v^\alpha.$$

$$\boxed{M du^\alpha + dM u^\alpha + dm v^\alpha = 0.} \quad (20)$$

Note: $dM + dm \neq 0$ in general; rest mass is not conserved because propellant carries kinetic energy too.

(d) Differential mass equation

Evaluate (c) in the IRF at τ : there $u_{\text{IRF}}^\alpha = (c, \vec{0})$ and, by (b), $du_{\text{IRF}}^0 = 0$, so $du_{\text{IRF}}^\alpha = (0, du, 0, 0)$ with du the IRF spatial velocity increment. The propellant moves backward at speed v in the IRF:

$$v_{\text{IRF}}^\alpha = \gamma_v(c, -v, 0, 0).$$

Take the time component ($\alpha = 0$) of the conservation equation in IRF:

$$M \cdot 0 + dM \cdot c + dm \cdot \gamma_v c = 0.$$

Solve for dM :

$$dM = -\gamma_v dm.$$

Take the spatial component ($\alpha = 1$):

$$M du + dM \cdot 0 + dm \cdot (-\gamma_v v) = 0.$$

Solve for dm :

$$dm = \frac{M du}{\gamma_v v}.$$

Substitute into the time-component result:

$$dM = -\gamma_v \cdot \frac{M du}{\gamma_v v} = -\frac{M du}{v}.$$

Divide both sides by M :

$$\boxed{\frac{dM}{M} = -\frac{du}{v}.} \quad (21)$$

(e) M as a function of launch-frame speed w

Let w be the rocket's speed in the launch frame and du its IRF speed increment. Velocity addition for the rocket's new velocity in the launch frame:

$$w + dw = \frac{w + du}{1 + w du/c^2}.$$

Expand to first order in du :

$$w + dw = (w + du)(1 - w du/c^2) = w + du(1 - w^2/c^2).$$

Isolate du :

$$du = \frac{dw}{1 - w^2/c^2}.$$

Substitute into the result of (d):

$$\frac{dM}{M} = -\frac{1}{v} \frac{dw}{1 - w^2/c^2}.$$

Integrate from 0 to w at constant v :

$$\ln \frac{M(w)}{M(0)} = -\frac{1}{v} \int_0^w \frac{dw'}{1 - w'^2/c^2}.$$

Evaluate the integral (partial fractions, or the hint $\int \sec \theta d\theta$ with $w'/c = \sin \theta$):

$$\int_0^w \frac{dw'}{1 - w'^2/c^2} = \frac{c}{2} \ln \left| \frac{1 + w/c}{1 - w/c} \right|.$$

Substitute:

$$\ln \frac{M(w)}{M(0)} = -\frac{c}{2v} \ln \frac{1 + w/c}{1 - w/c}.$$

Exponentiate:

$$\frac{M(w)}{M(0)} = \left(\frac{1 + w/c}{1 - w/c} \right)^{-c/(2v)} = \left(\frac{1 - w/c}{1 + w/c} \right)^{c/(2v)}.$$

Therefore:

$$\boxed{M(w) = M(0) \left(\frac{1 - w/c}{1 + w/c} \right)^\alpha, \quad \alpha = \frac{c}{2v}.} \quad (22)$$

Photon-rocket limit $v = c$ gives $\alpha = 1/2$, the slowest possible decay; non-relativistic limit $w/c \ll 1$ reduces to $M(w) \approx M(0)e^{-w/v}$, the classical Tsiolkovsky equation.